ORDER FLOW

Software Functional Specification

& Implementation Plan

Derived from: Michael Valtos – Trading Order Flow | *Dale – Order Flow Trading Setups*
Dale – Volume Profile Insider's Guide | *Dale – VWAP Book*
orderflows.com YouTube Channel | *Web Research (NinjaTrader, Wikipedia, LiteFinance, OptFutures)*

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1. Executive Summary

This document is the master functional specification for a commercial-grade Order Flow Software engine, targeting NinjaTrader 8 as the first platform with a defined roadmap to Sierra Chart, TradingView, Bookmap, and a standalone web dashboard.

The specification is synthesised from four authoritative source documents — Michael Valtos's Trading Order Flow, Dale's Order Flow Trading Setups, Dale's Volume Profile Insider's Guide, and Dale's VWAP Book — supplemented by the orderflows.com YouTube channel content and current industry web research including NinjaTrader's official order flow documentation, Wikipedia's order flow trading definition, and GoCharting's Orderflow Trader platform guide.

Key Product Goals

- Provide real-time, tick-by-tick analysis of bid/ask volume, delta, imbalance, absorption, and VWAP as described in the source books
- Surface institutional activity patterns invisible to standard candlestick or indicator-based analysis
- Deliver a configurable, extensible plugin API so users can build custom indicators on top of the engine's outputs
- Operate as a commercial subscription product with hardware-bound licensing and platform adapters
- Ship v1.0 on NinjaTrader 8 within 10–12 weeks, then scale to five additional platforms

2. Official Standards & Industry Definitions

There is no single ISO or regulatory body standard document for Order Flow trading software. However, the following de facto industry definitions and references constitute the accepted standard used by all major platforms (NinjaTrader, Sierra Chart, Bookmap, GoCharting, ClusterDelta):

2.1 Core Definitional Standard (Industry Consensus)

Term	Industry-Standard Definition	Source
Order Flow	Real-time analysis of executed market orders — specifically bid-side and ask-side volume — to determine market aggression and directional pressure	Wikipedia; NinjaTrader; Valtos
Footprint Chart	A chart that displays buy and sell volume executed at each individual price level within a bar. NinjaTrader calls these 'Volumetric Bars'. Industry standard term is Footprint Chart	NinjaTrader Official Blog 2025; TradeThePool
Delta	Net difference between ask-side volume (aggressive buyers) and bid-side volume (aggressive sellers) for a given bar. Positive = net buying; Negative = net selling	Valtos; Dale; LiteFinance
Cumulative Delta (CVD)	Running sum of delta values throughout the trading session. Tracks the cumulative difference between aggressive buyers and sellers across all bars	Dale; GoCharting Orderflows Trader Guide
Imbalance	When ask volume is 250%–400% greater than bid volume (buying imbalance) or bid volume is 250%–400% greater than ask volume (selling imbalance) at a diagonal price level comparison	Valtos; Dale; OptFutures
Stacked Imbalance	Three or more consecutive imbalances of the same type stacked vertically — acts as strong support or resistance zones	Valtos; Dale; OptFutures
Volume Profile	Histogram showing the distribution of traded volume across price levels for a given period. Key levels: Point of Control (POC), Value Area High (VAH), Value Area Low (VAL), Value Area = 70% of traded volume	Dale; LiteFinance
VWAP	Volume Weighted Average Price — average price weighted by volume. Formula: VWAP = $\Sigma(\text{Price} \times \text{Volume}) / \Sigma(\text{Volume})$. Widely used by institutional algorithms as benchmark	Dale VWAP Book; Citadel CEO Congressional testimony
Absorption	A price level where large volume trades on both bid and ask simultaneously, with minimal price movement — indicates institutional inventory transfer	Valtos; Dale
Unfinished Business	A high/low where the bid/ask at the extreme price	Dale;

(Failed Auction)	is not zero — the auction ended without proper completion, creating a price magnet for future revisit	ValtosTorderflows.com webinar 30
POC	Point of Control — the price level with the highest traded volume within a profile period. Most traded price represents institutional consensus of fair value	Dale; GoCharting
Value Area	The price range containing 70% of total traded volume for a period (VAH = upper boundary, VAL = lower boundary)	Dale; LiteFinance; GoCharting

2.2 Reference Sources Used

Source	Type	Coverage
Michael Valtos – Trading Order Flow (2015)	Primary reference book	Footprint charts, Delta, COT, Imbalances, Absorption, Trapped traders, Aggressive/Passive participants
Dale – Order Flow Trading Setups (2024)	Primary reference book	Volume Clusters, HVN, Cumulative Delta, Stacked Imbalances, Unfinished Business, Trade Filter, Confirmation setups
Dale – Volume Profile Insider's Guide (2024)	Primary reference book	Volume Profile shapes (D/P/b/Thin), POC, Value Area, Price Action strategies
Dale – VWAP Book (2024)	Primary reference book	VWAP calculation, Anchored VWAP, VWAP strategies, VWAP + Order Flow confluence
orderflows.com YouTube (@orderflows)	Video content / training	30+ webinar topics: Bar Delta Divergence, 2nd Slot Imbalances, Valtos U-Turn, Chart types, Iceberg orders, all Orderflows Trader 7 features
NinjaTrader Official Blog 2025	Platform standard	Footprint chart definition, Volumetric bars, Order Flow+ suite specification
Wikipedia – Order Flow Trading	Encyclopaedic reference	Buy/sell imbalance definitions, spoofing, DOM
GoCharting Orderflows Trader Guide	Platform implementation	Full feature list of Orderflows Trader indicators for automation
Orderflows.com Trader 7 spec	Commercial product spec	Complete list of 30+ automatable order flow signals from orderflows.com

3. Data Model & Core Data Structures

All order flow metrics derive from a single tick data stream. The following is the canonical data model.

3.1 Primary Tick Record

Field	Type	Description
timestamp_ns	uint64	Nanosecond-precision exchange timestamp
price	double	Executed price (tick precision)
volume	uint32	Number of contracts/shares at this tick
side	enum{BID, ASK}	Trade classification: ASK = aggressive buyer, BID = aggressive seller (Lee-Ready or exchange-flagged)
exchange_seq_no	uint64	Exchange sequence number for ordering
instrument_id	uint32	Internal instrument identifier

3.2 Price Level Record (Per Bar)

Field	Type	Description
price	double	Price level
bid_vol	uint32	Volume traded on bid side (aggressive sellers) at this price
ask_vol	uint32	Volume traded on ask side (aggressive buyers) at this price
total_vol	uint32	bid_vol + ask_vol
delta	int32	ask_vol - bid_vol (positive = net buying)
is_imbalance_buy	bool	ask_vol / bid_vol >= imbalance_ratio (default 3.0)
is_imbalance_sell	bool	bid_vol / ask_vol >= imbalance_ratio (default 3.0)

3.3 Bar Record

Field	Type	Description
open / high / low / close	double	Standard OHLC prices
bar_delta	int32	Sum of all level deltas in the bar
max_delta	int32	Maximum delta seen within the bar
min_delta	int32	Minimum delta seen within the bar
cumulative_delta	int64	Session cumulative delta up to and including this bar

total_volume	uint64	Total contracts traded in the bar
cum_volume	uint64	Session cumulative volume
poc_price	double	Price level with highest total_vol in this bar (bar POC)
cot_price	double	Commitment of Traders price = price with highest combined bid+ask volume
is_bullish_bar	bool	close > open
bar_type	enum{TIME, VOLUME, RANGE}	Chart type this bar belongs to
price_levels	vector<PriceLevelRecord>	All price levels within this bar

4. Feature Specifications

All features below are derived directly from the source documents and research. Each feature includes its source reference, calculation definition, display specification, and configurable parameters.

4.1 Footprint Chart (Core Visualisation)

Source: Valtos ch. 'Reading an Order Flow Chart'; Dale ch. 'Footprints'; NinjaTrader official blog; Dale ch. 'Basic Chart Description'

4.1.1 Bid/Ask Volume Display

- Each bar displays bid volume (left column) and ask volume (right column) at every price level inside the bar
- Bid column represents: aggressive sellers (sold into bid) + passive buyers (limit orders filled on bid)
- Ask column represents: aggressive buyers (bought from ask) + passive sellers (limit orders filled on ask)
- Cells are compared diagonally (ask at price N vs bid at price N-1) for imbalance detection — not horizontally
- Color coding: GREEN cell = ask_vol > bid_vol; RED cell = bid_vol > ask_vol; BLACK = neutral/balanced
- Volume shading: darker cell = higher volume (visual heat map to identify High Volume Nodes instantly)

4.1.2 Configurable Parameters

Parameter	Default	Range	Description
imbalance_ratio	3.0	1.5 – 10.0	Multiplier for imbalance detection (2.5x, 3x, 4x are industry standards)
min_vol_threshold	1	0 – 1000	Minimum volume to display in a cell (filters noise)
bar_type	TIME	TIME / VOLUME / RANGE	Chart construction method (Valtos recommends RANGE > VOLUME > TIME)
bar_size	5 min	1 min – monthly	Time period per bar (for TIME type)
range_size	4 ticks	1 – 100 ticks	Price range per bar (for RANGE type)
volume_size	1000	100 – 1M	Contracts per bar (for VOLUME type)
color_buy_cell	#00AA00	Any hex	Color for bullish (ask > bid) cells
color_sell_cell	#CC0000	Any hex	Color for bearish (bid > ask) cells

4.2 Delta Engine

Source: Valtos ch. 'The Importance of Delta'; Dale ch. 'Delta'; Wikipedia; GoCharting

- Bar Delta: ask_vol_sum - bid_vol_sum across all price levels in the bar
- Max Delta: highest delta value observed at any point within the bar construction
- Min Delta: lowest (most negative) delta value observed within the bar
- Cumulative Delta (CVD): running session total of bar deltas, reset at session open
- Delta/Volume %: bar_delta / bar_total_volume × 100 — normalises delta for different bar sizes
- Extreme Delta: flags bars where abs(delta) exceeds a configurable extreme_delta_threshold
- Small Min/Max Delta: flags bars where range between min_delta and max_delta is unusually small (consolidation indicator)

4.2.1 Delta Divergence Detection (Valtos / Dale)

- Standard Delta Divergence: negative delta at a new/equal high OR positive delta at a new/equal low
- Bar Delta Divergence (Valtos Webinar 18): red (bearish) bar with positive delta OR green (bullish) bar with negative delta — go in direction of the candle color
- CVD Divergence (Dale): price making new highs while cumulative delta is falling, or price making new lows while CVD is rising — indicates imminent reversal

📌 *NOTE: Delta divergence is one of the highest-conviction signals in order flow analysis per both Valtos and Dale. Must be detected in real time and flagged for user.*

4.3 Imbalance Detection

Source: Valtos ch. 'Imbalance Levels'; Dale ch. 'Imbalances'; OptFutures; LiteFinance

4.3.1 Single Imbalance

- Buying Imbalance: ask_vol[N] / bid_vol[N-1] >= imbalance_ratio (diagonal comparison upward)
- Selling Imbalance: bid_vol[N] / ask_vol[N+1] >= imbalance_ratio (diagonal comparison downward)
- Display: highlight imbalance cell in blue (buying) or blue (selling) as per Dale; Valtos uses green/red zones
- Industry standard ratios: 250% (2.5x), 300% (3x), 400% (4x) — user configurable

4.3.2 Stacked Imbalance (Critical Feature)

- Definition: 3 or more consecutive imbalances of the same type vertically stacked
- Stacked buying imbalance = strong demand zone (support); stacked selling imbalance = strong supply zone (resistance)
- Display: draw coloured zone rectangle on chart (green for buy stacks, red for sell stacks)

- Zone persists on chart across subsequent bars as support/resistance reference
- Stop placement reference: 1–2 ticks outside the stacked imbalance zone (Valtos rule)
- Entry reference: limit order at middle of stacked imbalance zone (Valtos rule)

4.3.3 Additional Imbalance Types

Imbalance Type	Definition	Source
2nd Slot Imbalance	Imbalance appearing 1–2 price levels away from the initial imbalance zone — late-entry traders re-joining the move	Valtos Webinar 19
Imbalance Reversal	Imbalance that appears at the opposite direction after an exhaustion — signals trend reversal zone	Orderflows Trader 7 spec
Multiple Imbalances	Cluster of imbalances across multiple bars at same price — institutional accumulation zone	Orderflows Trader 7 spec
Inverse Imbalance	Imbalance occurring in the opposite direction to the prevailing trend — early reversal warning	Orderflows Trader 6 spec
Imbalance Reload	A second imbalance appearing at the same price level in a subsequent bar — strong re-entry confirmation	Orderflows Trader 7 spec

4.4 Volume Profile Engine

Source: Dale Volume Profile book; Valtos ch. 'The Importance of Volume'; Dale Order Flow setups ch. 'Volume Profile'; LiteFinance

4.4.1 Profile Types

Profile Type	Period	Use Case
Session Profile	Single trading session (e.g., 9:30am–4:00pm)	Intraday trading reference — most common
Daily Profile	One calendar day	Day-over-day analysis
Weekly Profile	Monday–Friday	Swing trade context
Monthly Profile	Calendar month	Institutional positioning analysis
Flexible / Anchored Profile	User-defined start to current bar	Custom range analysis (Dale's preferred method)
Composite Profile	Merged across multiple sessions	Long-term supply/demand mapping

4.4.2 Profile Key Levels

- Point of Control (POC): price level with highest total volume — most traded price, institutional consensus of fair value

- Value Area High (VAH): upper boundary of the value area (70% of volume)
- Value Area Low (VAL): lower boundary of the value area (70% of volume)
- Value Area (VA): price range from VAL to VAH containing 70% of session volume
- High Volume Node (HVN): price level where volume significantly exceeds surrounding levels — strong support/resistance
- Low Volume Node (LVN): price level with very low volume — price tends to move quickly through these zones

4.4.3 Profile Shapes (Dale's classification)

Shape	Visual Pattern	Market Interpretation
D-shaped (Bell curve)	Wide symmetric profile centered on POC	Balanced, rotational market. Two-sided trade. No strong directional bias.
P-shaped	Narrow base, wide top — POC near high	Bullish accumulation at lows, distribution near highs. Look for continuation up or short at rejection.
b-shaped	Wide base, narrow top — POC near low	Bearish distribution. Selling near highs, accumulation near lows. Look for continuation down.
Thin Profile	Narrow, elongated profile — minimal volume	Trending/initiative market. Price moving quickly through area. Low confidence in these levels as support/resistance.

4.4.4 Prominent POC & Aligned POCs

- Prominent POC: a POC that stands significantly above surrounding volume — highest-conviction institutional level (Orderflows Trader 7)
- Aligned POCs: two or more POCs aligned at the same price across consecutive sessions — extremely strong level of support/resistance (Orderflows Trader 7)
- Open POC: the current session's POC as it is building in real time (Orderflows Trader 7)
- POC Wave: the path traced by POC migration during session construction — shows institutional intent direction
- POC Slingshot (Valtos): when price breaks away from POC with strong delta — high-probability continuation trade

4.5 VWAP Engine

Source: Dale VWAP Book; Citadel CEO Congressional Testimony (govinfo.gov); Valtos ch. 'VWAP'

4.5.1 VWAP Calculation

- Formula: VWAP = Σ(Price_i × Volume_i) / Σ(Volume_i) [cumulative from anchor point]
- VWAP is superior to SMA/EMA because it incorporates volume — reflects where the average trader actually transacted

- Institutional use: virtually all institutional orders are executed as VWAP program trades (Citadel CEO testimony, 2021)

4.5.2 VWAP Types

VWAP Type	Anchor Point	Preferred Timeframe	Use Case
Daily VWAP	Session open (Asian session start)	5 min	Intraday. Most widely used. Support/resistance for day traders.
Weekly VWAP	Monday Asian session open	30 min	Intraday + swing. Dale's personal preference for reliable signals.
Yearly VWAP	January 1 session open	Daily	Swing + long-term. Fair price benchmark for the year.
Anchored VWAP – Swing Point	Key swing high or low	Any	Trade pullbacks to VWAP from pivotal market turning points
Anchored VWAP – Macro News	Candle of major news event	Any	Measures average price since a regime-changing event
Anchored VWAP – Heavy Volume	Start of a volume cluster	Any	Tracks fair value since institutional accumulation/distribution began
Anchored VWAP – Gap	Pre-gap close / post-gap open	Any	Measures fill progress and fair value relative to gap event
Anchored VWAP – Earnings	Earnings announcement candle	Daily	Stock-specific fair value since earnings report

4.5.3 VWAP Strategy Rules (from Dale)

- If price is above VWAP: VWAP acts as support — look for long entries on pullbacks to VWAP from above
- If price is below VWAP: VWAP acts as resistance — look for short entries on rallies to VWAP from below
- VWAP break and hold: if price convincingly crosses VWAP and stays, role reversal — former support becomes resistance and vice versa
- VWAP + Order Flow confluence: most powerful entry when Order Flow confirms aggression at VWAP level (Valtos + Dale both use this)

4.5.4 VWAP Standard Deviations

- 1st Standard Deviation band (±1σ): most common mean-reversion zone
- 2nd Standard Deviation band (±2σ): extreme price zone — VWAP rotation strategy target
- VWAP Rotation Strategy (Dale): enter long at -1σ targeting VWAP; enter short at +1σ targeting VWAP
- VWAP Trend Strategy (Dale): price breaks and holds above +1σ — trend continuation, enter pullbacks to +1σ

4.6 Commitment of Traders (COT) — Intrabar Version

Source: Valtos ch. 'COT — Commitment of Traders'

📌 *NOTE: This is NOT the CFTC COT report. This is Valtos's proprietary intrabar COT concept.*

- COT Price: the price level within a bar where the highest combined bid + ask volume occurred
- Display: draw a rectangle/box around the COT price level for easy visualisation
- COT near low of bar, bar closes high: strong passive buyers — bullish signal
- COT near high of bar, bar closes low: strong passive sellers — bearish signal
- COT in middle of bar: neutral — market facilitation, no strong directional signal
- COT use: confirmation tool when market is range-trading or direction is unclear

4.7 Trapped Buyers & Sellers Detection

Source: Valtos ch. 'Trapped Buyers and Sellers'

- Trapped Sellers: selling imbalances appear at or near the LOW of a bar that subsequently closes HIGHER — sellers who sold the low are now trapped short
- Trapped Buyers: buying imbalances appear at or near the HIGH of a bar that subsequently closes LOWER — buyers who bought the high are now trapped long
- Trapped traders must eventually exit (cover) their losing position, adding fuel to the move against them
- Detection rule: imbalances within 2 ticks of bar high/low where bar closes opposite to imbalance direction
- Signal: short-term reversal opportunity with good R/R — trapped traders' covering creates directional momentum

4.8 Absorption Detection

Source: Valtos ch. 'Absorption'; Dale ch. 'Confirmation Setup #2: Absorption'

- Absorption: large volume trades on BOTH bid and ask at a price level simultaneously, with net delta close to zero relative to total volume
- Mathematical definition: total_vol > absorption_vol_threshold AND abs(delta) / total_vol < absorption_delta_ratio

- Default thresholds: absorption_vol_threshold = 500 contracts; absorption_delta_ratio = 0.1 (delta is <10% of total vol)
- Market interpretation: institutional inventory transfer — big money changing hands without moving price
- Significance: marks potential trend reversal or breakout origin — once absorption ends, price is free to move
- Distinguish from low-volume consolidation: absorption requires HIGH volume at the level, not low volume
- Caveat (Valtos): end-of-day absorption may be day traders closing positions — filter by session context

4.9 Initiative vs. Responsive Activity

Source: Valtos ch. 'Initiative Activity'; Dale ch. 'Price Action'

- Initiative Activity: price moves ABOVE value area high (buyers taking initiative) or BELOW value area low (sellers taking initiative) with strong delta support
- Responsive Activity: price reaches an area far from value and passive participants push it back with limit orders
- Detection: compare current price to session VAH/VAL; measure delta at breakout — high delta breakout = initiative; low delta, quick reversal = responsive
- Trading implication: trade in direction of initiative activity; trade against price in responsive activity zones

4.10 Special Order Flow Signals (Orderflows Trader 7 / Valtos Webinars)

All signals below are from orderflows.com Trader 7 specification and Inner Circle webinar series, confirmed by research.

Signal Name	Definition	Trading Use
Market Sweep Detector	Detects when large aggressive orders sweep through multiple price levels in one burst — iceberg-style institutional entry	High-conviction trend entry — join the sweep direction
Market Weakness Detector	Detects bars where price reaches new high/low but volume/delta is declining — momentum exhaustion	Exit existing trade or prepare for reversal
Zero Prints (Thin Prints)	Price levels within a bar where ZERO contracts traded — extreme imbalance, unfinished auction	Price magnet for future revisit; defines tight stop zones
Exhaustion Print	Volume spike at the extreme of a move with delta reversal — final capitulation of one side	High-probability reversal entry
Orderflows Sequencing	Pattern of consecutive bars showing increasing or decreasing delta in same direction — trend strength measurement	Confirm trend continuation; exit when sequence breaks

Delta Tail (Delta Breakout)	Delta breaks above/below its recent range at a key price level — institutional order flow entering	Confirm breakout entries; validate level strength
Resting Liquidity	Significant volume sitting at a price without moving price — passive orders absorbing supply/demand	Identifies hidden support/resistance not visible on standard charts
Vertical Liquidity	Volume stacking vertically at consecutive price levels — large institution working a big order over multiple levels	Defines institutional entry zone
Accumulation/ Distribution	Detects multi-bar accumulation (repeated buying near lows) or distribution (repeated selling near highs)	Early signal of trend change before price action confirms
Orderflows Ratio	Ratio of ask volume to bid volume for the entire bar — normalised measure of directional pressure	Quantifies bar strength; filter for high-conviction bars
Orderflows Gaps	Volume gaps at price levels inside a bar — areas with no trades that price skipped through	Future support/resistance magnets; stop placement zones
Price Action Divergence (Valtos U-Turn)	Detects where buyers/sellers gave up and market reversed due to hidden supply or demand	Reversal entries; identifies hidden resistance/support
POC Slingshot	Price breaks away from POC with strong delta confirmation after a period of balance	High-probability trend-following entry
Volume Decline	Sequential decrease in volume across bars in same direction — trend running out of fuel	Exit or reduce position size; reversal watch
Engulfing Value Area	Current bar's range completely engulfs previous bar's value area — high-confidence breakout	Trend continuation or reversal depending on context
Prominent POC / Retail Suck	Price briefly spikes through a key level to run retail stops before reversing — institutional trap	Fade the spike; enter in original direction after reversal

5. Trading Setups — Functional Requirements

The engine must detect and optionally alert/annotate the following canonical trading setups drawn from all four source books.

5.1 Primary Order Flow Setups (Dale's Book)

Setup Name	Entry Condition	Confirmation	Stop Loss	Take Profit
Volume Cluster – Trend	Price pulls back into a high-volume cluster in direction of prevailing trend	Delta confirms; CVD not diverging	Below/above volume cluster	Next significant HVN or POC
Volume Cluster – Rejection	Price reaches a cluster at key level and shows aggressive rejection delta	Bar closes away from cluster strongly	Inside the cluster zone	Previous high/low, next VWAP level
Multiple Nodes (HVN)	Two or more HVNs align at same price across consecutive bars — yellow zone	Delta neutral or confirming at the zone	Clearly outside the yellow zone	POC or opposite value area boundary
Trades Filter	Large single-execution trades (25+ contracts) appear at key support/resistance level	Price reacts immediately to the large order	Opposite side of the large order	1:2 or better R/R to next structure
Stacked Imbalances	Price retraces into a stacked imbalance zone (3+ imbalances same direction)	Entry at middle of zone; stop 1–2 ticks outside	1–2 ticks outside stacked zone	Next imbalance zone opposite side; POC
Unfinished Business	Price has left a zero-print at previous high/low — magnet effect draws price back	CVD divergence at unfinished business level	Opposite side of the zero print	The unfinished business level itself (for continuation trades)

5.2 Confirmation Setups (Dale's Book)

Confirmation Type	Signal	Usage
Big Limit Orders	Large passive orders visible at price level absorbing market orders — COT location confirms	Confirms support/resistance; entry trigger when price reacts strongly

Absorption	High volume, near-zero net delta at key level	Confirms trend reversal or breakout origin point
Aggressive Orders + Delta	Strong delta in direction of setup at the entry level	Confirms momentum behind the trade setup
CVD Divergence	Price going one direction while CVD goes opposite	Highest-conviction reversal confirmation in Dale's methodology

5.3 VWAP Trading Setups (Dale's VWAP Book)

Setup	Entry	Confirmation	Target
VWAP Reaction	Price touches VWAP from direction of prevailing trend	Delta confirms in trend direction at VWAP; Order Flow shows absorption	Previous swing high/low; -1σ to +1σ
VWAP Rotation	Price at +1σ (short) or -1σ (long) with rotation signal	Delta divergence; stacked imbalance at deviation band	VWAP (center)
VWAP Trend	Price breaks and holds above +1σ; pullback to +1σ	Delta stays positive on pullback; CVD rising	Next VWAP deviation band above
Anchored VWAP + Volume Profile	Price at Anchored VWAP that also coincides with Volume Profile POC or VAH/VAL	Delta confirmation; absorption at the confluence zone	Opposite boundary of value area or next swing structure

5.4 Volume Profile Setups (Dale's Volume Profile Book)

Setup	Logic	Entry Trigger
Volume Accumulation	Identify D-profile (balanced) area. Wait for price to leave the range with strong delta. Trade breakout direction.	First bar closing outside value area with 3x normal delta
Trend Setup (P/b-profile)	P-profile = bullish trend setup; b-profile = bearish trend setup. Trade in profile direction.	Pullback to POC or VAH/VAL of the profile with delta confirmation
Rejection Setup	Price reaches extreme level and volume profile shows thin profile — fast rejection likely	High delta rejection bar at thin profile extreme; trade back toward value

6. External API Specification

The public API allows external users and custom indicator developers to consume all engine outputs without accessing the core C++ engine code. The API is read-only — no write access to the engine.

6.1 REST API Endpoints

Endpoint	Method	Description	Auth Required
/v1/instruments	GET	List all available instruments	API Key
/v1/bars/{instrument_id}	GET	Historical bars with all order flow metrics (delta, POC, imbalances, etc.)	API Key
/v1/profile/{instrument_id}	GET	Volume profile for given period (session/daily/weekly/monthly/anchored)	API Key
/v1/vwap/{instrument_id}	GET	VWAP levels (daily/weekly/yearly) with standard deviation bands	API Key
/v1/signals/{instrument_id}	GET	Active detected signals (stacked imbalances, unfinished business, absorption zones)	API Key
/v1/config	GET/ PUT	Read or update user configuration parameters	JWT Token
/v1/license/status	GET	Current license tier, expiry, and feature availability	JWT Token

6.2 WebSocket Streaming API

Stream Topic	Payload	Rate
ticks/{instrument}	Raw tick: {timestamp_ns, price, volume, side}	Every tick (realtime)
bars/{instrument}	Complete bar record on bar close: all OFE metrics	Per bar close
delta/{instrument}	Live bar delta as it builds: {bar_delta, cumulative_delta, bar_volume}	Every 100ms
signals/{instrument}	Signal detection events: {signal_type, price, strength, direction}	Event-driven
vwap/{instrument}	VWAP level updates: {vwap_daily, vwap_weekly, sd1_high, sd1_low, sd2_high, sd2_low}	Every 5 seconds
profile/{instrument}	Volume profile updates: {poc, vah, val, hvns[], lvns[]}	Every bar close

6.3 Custom Indicator Plugin Interface (C ABI)

External developers can implement custom indicators as shared libraries (.dll/.so) against the following C ABI:

- OFE_IndicatorInit(ctx, config_json) — initialise with engine context and JSON config
- OFE_OnTick(ctx, tick_record) — called on every new tick
- OFE_OnBarClose(ctx, bar_record) — called on every bar close with full bar data
- OFE_OnSignal(ctx, signal_type, payload) — called when engine detects a registered signal
- OFE_GetOutput(ctx, output_key) — returns current indicator output value for display
- OFE_Destroy(ctx) — cleanup

7. User Configurability Requirements

Per Valtos's design philosophy: 'I purposely wanted to keep the chart clean and uncluttered so that it is easy to see setups as they are happening.' All parameters below are user-configurable via YAML config file with hot-reload (no restart required).

7.1 Global Configuration Parameters

Category	Parameter	Default	Description
Chart	bar_type	TIME	TIME / RANGE / VOLUME chart type
Chart	bar_size	5min	Size of each bar for selected bar_type
Chart	show_session_hilo	true	Show session high/low line (Valtos's preferred minimal chart element)
Delta	imbalance_ratio	3.0	Ratio for imbalance detection (2.5, 3.0, 4.0)
Delta	stacked_min_count	3	Minimum consecutive imbalances to classify as 'stacked'
Delta	delta_divergence_enabled	true	Enable/disable delta divergence detection
Volume Profile	value_area_percent	70	Percentage of volume defining the value area (standard = 70%)
Volume Profile	profile_type	SESSION	SESSION / DAILY / WEEKLY / MONTHLY / FLEXIBLE
Volume Profile	show_poc	true	Display Point of Control
Volume Profile	show_hvn	true	Display High Volume Nodes
VWAP	vwap_types	DAILY, WEEKLY	Which VWAP types to calculate and display
VWAP	vwap_sd_bands	1,2	Which standard deviation bands to display (1σ, 2σ)
Signals	trades_filter_size	25	Minimum contracts for Trades Filter signal (Dale: 25 for EUR futures)
Signals	absorption_vol_threshold	500	Minimum contracts for absorption detection
Signals	absorption_delta_ratio	0.10	Max delta/volume ratio to classify as absorption
Display	color_scheme	DARK	DARK / LIGHT / CUSTOM
Display	cell_min_font_size	7	Minimum font size for cell volume numbers (readability)

API	api_rate_limit_per_min	1000	API calls per minute for this subscription tier
Alerts	signal_alert_enabled	true	Enable desktop/sound alerts for signal detection

8. Implementation Plan

8.1 Architecture Overview

The system is built as a C++ core engine with thin platform adapters. All business logic lives in the engine — platform adapters contain zero business logic.

Layer	Technology	Responsibility
Tick Ingestion	C++ lock-free SPSC ring buffer	Receive raw tick stream from data provider at nanosecond precision
Trade Classification	C++ Lee-Ready algorithm	Classify each tick as bid-side or ask-side aggression
Bar Construction	C++ time/volume/range bar engine	Aggregate ticks into bars per configured bar_type
Order Flow Analytics	C++ analytics engine	Calculate all metrics: delta, CVD, imbalances, absorption, COT, traps
Volume Profile	C++ profile engine	Build and maintain all profile types (session/daily/weekly/flexible)
VWAP Engine	C++ VWAP calculator	Calculate and maintain all VWAP types including anchored variants
Signal Detection	C++ pattern engine	Detect all 30+ signals from the feature spec in real time
Storage	Memory-mapped files + SQLite	Persist tick history for replay and historical analysis
REST API	C++ with cpp-httplib or Crow	Serve historical data and config to external consumers
WebSocket API	C++ with uWebSockets	Stream real-time analytics to external consumers
NinjaTrader Adapter	C# P/Invoke wrapper	Thin bridge calling C++ DLL — zero business logic
License Engine	C++ with RSA-2048	Hardware fingerprint, online validation, subscription enforcement
Config System	C++ YAML parser + hot-reload	Load and apply user configuration without restart

8.2 Development Phases

Phase	Timeline	Deliverables
Phase 0: Interfaces	Week 1	All C++ header files (.h) for every module — approved before any implementation begins
Phase 1: Core Engine	Week 2–4	Tick ingestion, bar construction, delta engine, basic imbalance detection, footprint display data

Phase 2: Analytics	Week 5–6	Full imbalance suite (stacked, 2nd slot, imbalance reversal), absorption, trapped traders, COT, initiative activity
Phase 3: Profile + VWAP	Week 7–8	Volume profile engine (all types + shapes), VWAP engine (all types + std dev bands), POC logic
Phase 4: Signal Engine	Week 9–10	All 30+ signal detections, alert system, Orderflows Trader 7 full signal set
Phase 5: API Layer	Week 11–12	REST API, WebSocket streaming, Plugin C ABI, API key management
Phase 6: NinjaTrader	Week 13–14	C# P/Invoke adapter, NinjaTrader workspace integration, NinjaTrader-specific display rendering
Phase 7: License + Security	Week 15–16	Hardware fingerprinting, license server backend, binary obfuscation (Themida/VMProtect), anti-tamper
Phase 8: Testing	Week 17–20	Unit tests, historical tape replay, synthetic stress test, live paper trading beta with 10 users
Phase 9: v1.0 Launch	Week 21–24	NinjaTrader public release, subscription system live (Paddle), documentation, support
Phase 10: Sierra Chart	Month 7–9	ACSIL C++ study adapter for Sierra Chart — same C++ engine, new thin wrapper
Phase 11: Web Dashboard	Month 10–14	WebSocket-based web UI (React), accessible via browser — v2.0 release
Phase 12: TradingView	Month 12–16	WebSocket bridge to TradingView Pine Script; REST API for historical
Phase 13: Bookmap + TT	Month 14–18	JNI bridge for Bookmap; TT SDK adapter for Trading Technologies

8.3 Testing Strategy

Layer	Method	Criteria
Unit Tests	Google Test (gtest) — all calculation modules	100% pass on known inputs with verified expected outputs
Historical Replay	Databento or Rithmic historical tick data — full session replay	All metrics match reference implementation within float precision
Synthetic Stress	Synthetic tick generator: 500,000 ticks/sec, 0.1% out-of-order timestamps	Zero corruption of delta/CVD calculations; ring buffer no overflow
Signal Validation	Replay known setups from source books with known outcomes	All 6 primary setups + 4 confirmations detected correctly
API Load Test	1000 concurrent WebSocket connections, 10 REST requests/sec each	P99 latency < 50ms; zero data corruption
NinjaTrader Integration	Live paper account on NinjaTrader	Visual output matches expected

	8 with Rithmic data feed	footprint charts from Valtos book
License System	Tamper simulation; hardware clone attempt; expiry enforcement	License invalid on cloned hardware; engine shuts down on expiry

8.4 Agent Pipeline Execution Order (Claude Code)

📌 *NOTE: For implementation using the Claude Code agent pipeline discussed in prior sessions.*

Agent	Inputs	Outputs	Human Review Gate
Arch Agent	This spec document + NinjaTrader addon documentation	Full architecture doc, UML diagrams, interface contracts	Review architecture before any code
Interface Agent	Architecture doc	All C++ .h header files — zero implementation	CRITICAL: approve all headers before code generation
Code Agent (Core)	Approved headers	Tick ingestion, bar construction, delta engine implementation	Review for correctness and thread safety
Code Agent (Analytics)	Core code + approved headers	Imbalance, absorption, COT, trapped traders, signals	Review formulas against spec
Code Agent (Profile/VWAP)	Approved headers	Volume profile engine, VWAP calculator	Validate against Dale's book formulas
Code Agent (API)	Approved headers	REST + WebSocket API, plugin C ABI	Review API contracts match spec
Test Agent	All code	gtest unit tests, synthetic generator, replay harness	Run tests, fix all failures
Security Agent	Compiled binary	License engine, anti-tamper integration	Manual: Themida/VMProtect configuration
Review Agent	All code	Final code review report, gap list	Sign off before beta release

9. Complete Glossary

All terms derived from source books and web research.

Term	Definition
Absorption	Large volume on both bid and ask at a level with minimal price movement — institutional inventory transfer
Aggressive Order	Market order — buyer or seller who pays the spread to transact immediately
Anchored VWAP	VWAP whose calculation begins at a specific user-defined anchor point rather than session open
Bar Delta	Net ask_vol - bid_vol for a single bar
Bar Delta Divergence	Bearish bar (red close) with positive delta, or bullish bar (green close) with negative delta — trade candle direction
COT (Intrabar)	Valtos's Commitment of Traders: price level of highest combined bid+ask volume within a bar
Cumulative Delta (CVD)	Session running total of all bar deltas
Delta	Net difference between aggressive buying volume (ask) and aggressive selling volume (bid)
D-Profile	Bell-curve volume profile shape — balanced, rotational market
Exhaustion Print	Volume spike at move extreme with delta reversal — final capitulation
Failed Auction	See Unfinished Business
Footprint Chart	Chart showing bid and ask volume at each price level within each bar
High Volume Node (HVN)	Price level with significantly higher volume than surrounding levels — institutional consensus zone
Iceberg Order	Large institutional order broken into small pieces to avoid market impact
Imbalance	Ask vol ≥ N × bid vol (buy imbalance) or bid vol ≥ N × ask vol (sell imbalance) at diagonal price comparison
Initiative Activity	Price moving outside the value area with strong delta — institutional trend-starting activity
Lee-Ready Algorithm	Standard academic tick classification algorithm for determining if a trade was buyer or seller-initiated
Low Volume Node (LVN)	Price level with very low volume — price moves quickly through these zones
Passive Order	Limit order — buyer or seller waiting at a price for the market to come to them
POC	Point of Control — price with highest total traded volume in a profile period
P-Profile	Volume profile shape with wide top and narrow base — bullish

	accumulation
b-Profile	Volume profile shape with wide base and narrow top — bearish distribution
Responsive Activity	Limit orders pushing price back toward fair value from an extreme
Stacked Imbalance	3+ consecutive imbalances of same type — strong support/resistance zone
Standard Deviation (VWAP)	Statistical measure of price dispersion from VWAP — used to define trading bands
Thin Profile	Narrow elongated volume profile — trending/initiative market
Trapped Buyers/Sellers	Traders who bought at high/sold at low with price now moving against them — must cover, fuelling the move
Unfinished Business	High or low where bid/ask at the extreme is not zero — price magnet for future revisit
Value Area	Price range containing 70% of session volume (between VAL and VAH)
VWAP	Volume Weighted Average Price — Σ(price × vol) / Σ(vol) — institutional benchmark price
Zero Print	Price level within a bar where zero contracts traded — same as thin print / unfinished auction extreme

END OF SPECIFICATION DOCUMENT — v1.0

Sources: Valtos (2015), Dale (2024 ×3), orderflows.com, NinjaTrader, Wikipedia, GoCharting, LiteFinance, OptFutures